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System Flowchart Documentation

DYS Sales Tracker Management System

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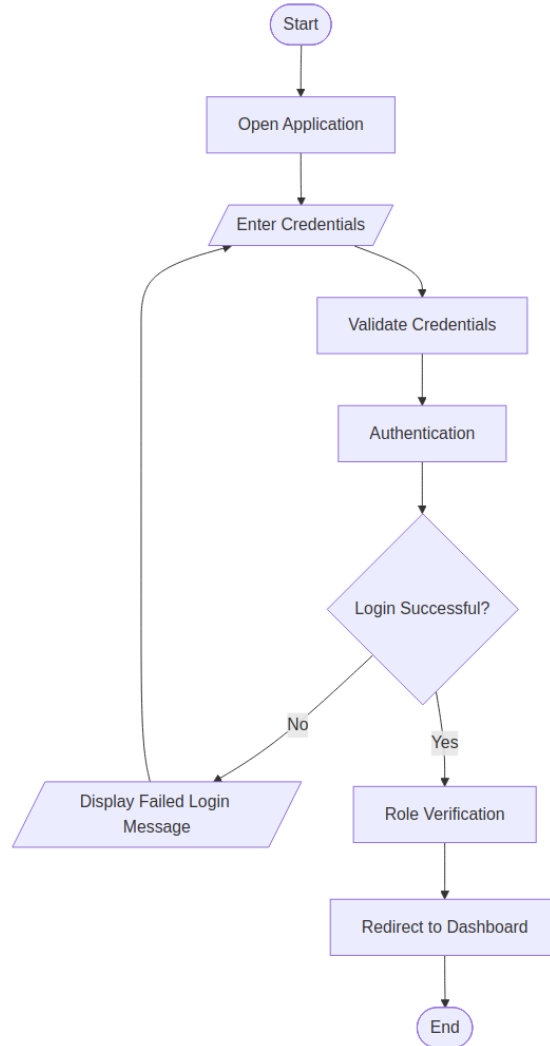
Introduction

The system flowcharts presented in this section describe how the DYS Sales Tracker Management System is expected to behave for each of its major business processes. Since the application is meant to replace a mostly manual method of recording sales, expenses, and related financial information, it was necessary to lay out these processes clearly before any part of the system was built, so that the intended behavior could be checked against the requirements described in the approved Concept Paper.

Five processes are documented here, corresponding to the major features approved for the system: logging in, recording a sales transaction, recording an expense, generating a report, and switching between the business sectors managed by the client. Each diagram traces the sequence of actions carried out by the user and by the system for a single process, including the points at which the system must evaluate a condition before it can continue, such as confirming a login attempt or checking that an amount entered by the user is valid. The diagrams use a small, consistent set of symbols throughout: an oval marks the start or end of a process, a rectangle represents an action carried out by the system, a parallelogram represents information supplied by the user or shown back to them, and a diamond represents a decision that determines which path the process takes next.

Flowchart 1 — Login Process

To verify the identity of a user attempting to access the system and to direct that user to the dashboard associated with their assigned role.



Flowchart 1 — Login Process diagram

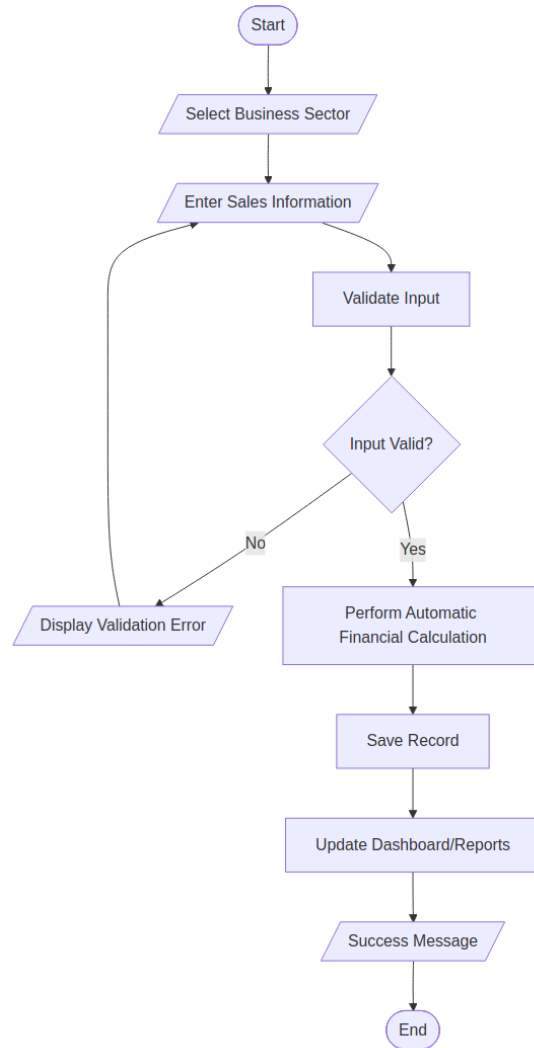
Process Description

The login process begins when a user opens the DYS Sales Tracker mobile application and is presented with the login screen. The user is asked to enter a username and password, which the system checks for completeness before attempting to authenticate the account against the records stored on the server. Because the system supports three types of users — the Business Owner, the Event Manager, and Employees or Event Staff — a successful login does not by itself determine what a user is allowed to see or do; the system must still confirm which role has been assigned to that account.

If the credentials entered do not match an existing account, the user is informed that the login attempt failed and is returned to the login screen to try again. If the credentials are correct, the system identifies the user's role and opens the dashboard that corresponds to it, so that a Business Owner, an Event Manager, and a member of the event staff are each shown only the functions appropriate to their position. This process establishes the starting point for the role-based access control required throughout the rest of the system.

Flowchart 2 — Record Sales Process

To allow an authorized user to record a sales transaction under the correct business sector and have the corresponding totals updated without manual calculation.



Flowchart 2 — Record Sales Process diagram

Process Description

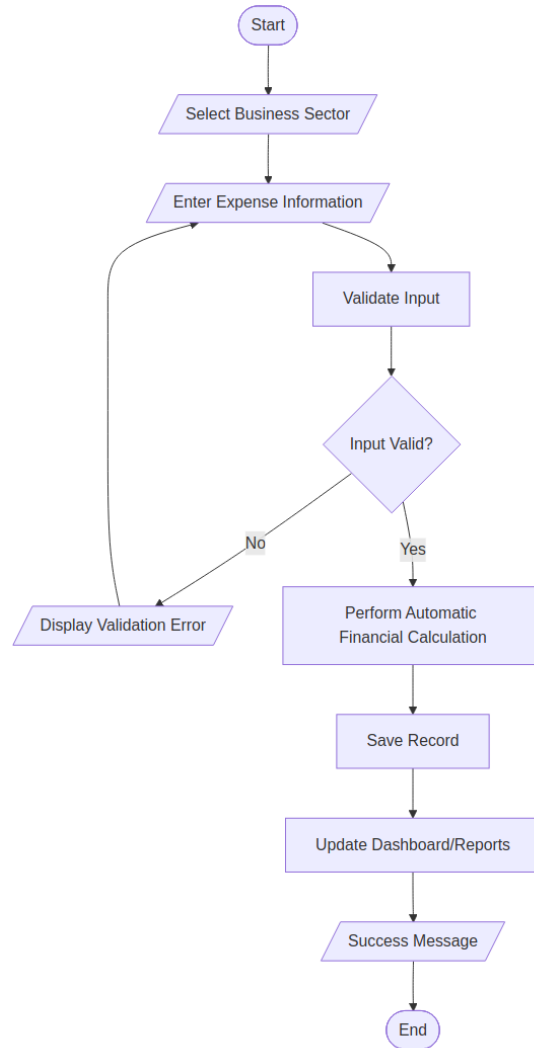
Recording a sales transaction begins with the user identifying which business sector the transaction belongs to, since the records of each sector are kept separate within the system. Once the sector has been selected, the user enters the details of the sale, and the system checks this information before it is accepted. Entries that are incomplete or contain an invalid value are rejected, and the user is asked to correct them before continuing.

Once the entered information passes validation, the system performs the calculation associated with the automated financial calculator, updating the running totals for that sector without requiring the user to compute anything manually. The transaction is then saved, the dashboard and related reports

are updated to reflect the new figures, and the user is shown a message confirming that the record was stored.

Flowchart 3 — Record Expenses Process

To allow an authorized user to record a business expense under the correct business sector and have the corresponding totals updated automatically.



Flowchart 3 — Record Expenses Process diagram

Process Description

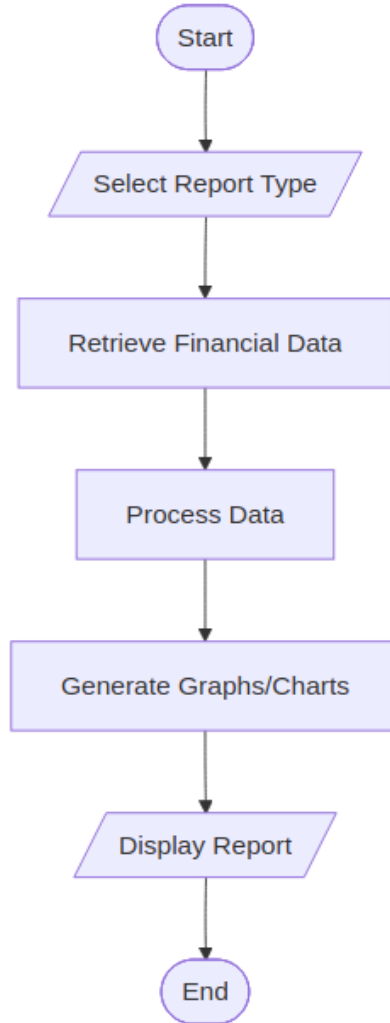
The process for recording an expense follows the same general structure as recording a sale, since both are handled by the same underlying financial calculator. The user first selects the business sector to which the expense applies, then supplies details such as the amount and nature of the expense. The system validates this information before it is stored, returning the user to the entry screen with an explanation if something is missing or entered incorrectly.

Valid entries are used to update the sector's expense totals automatically, after which the record is saved and reflected in the dashboard and reports for that sector. A confirmation message lets the user know that the expense has been logged. Keeping this process nearly identical to the sales

process was a deliberate choice, since both types of transactions affect the same financial totals and should be handled consistently by the system.

Flowchart 4 — Generate Reports Process

To convert the sales and expense records stored in the system into visual summaries that support the Business Owner and Event Manager in reviewing performance.



Flowchart 4 — Generate Reports Process diagram

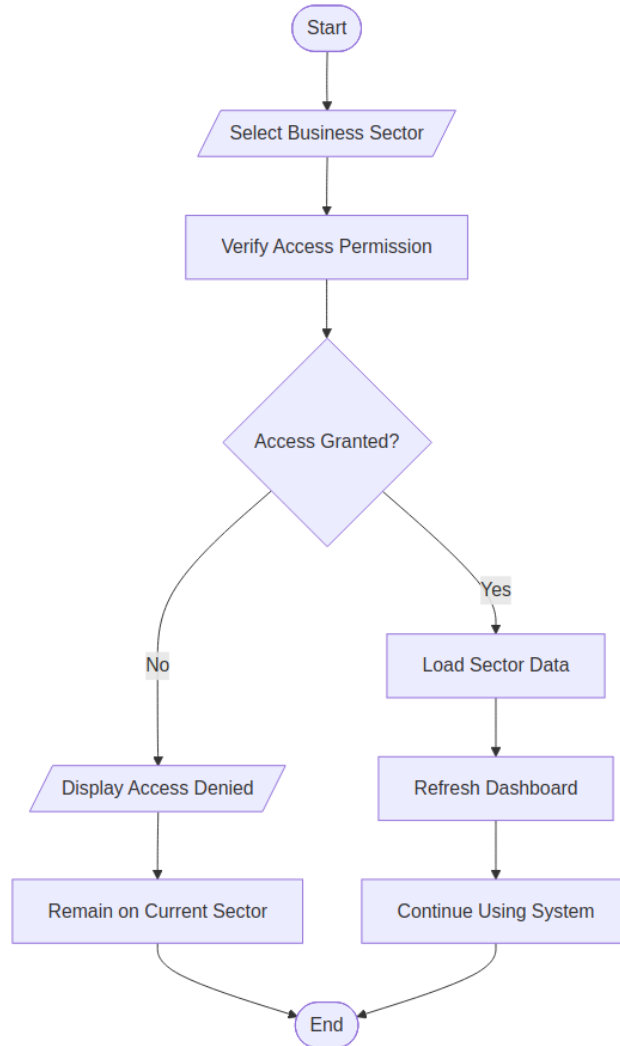
Process Description

Unlike the other processes documented here, generating a report does not involve a decision point, since the process is meant to run in a straightforward manner once a report type has been selected. After the user chooses the type of report to view, the system retrieves the relevant sales and expense records for the active business sector and processes this information to produce totals and trends.

The processed data is then presented to the user as a chart or graph, allowing performance to be reviewed at a glance rather than through raw figures. Because this process only reads existing records and does not create or modify any data, it does not require the same kind of error handling used in the processes that involve data entry.

Flowchart 5 — Switch Business Sector Process

To allow an authorized user to move between the business sectors managed within the system while keeping the records of each sector properly separated.



Flowchart 5 — Switch Business Sector Process diagram

Process Description

When a user selects a different business sector from the dashboard, the system first checks whether that user has permission to access the sector in question. This step exists because not every user is expected to have the same level of access to every part of the business, and the system needs to prevent someone from viewing or modifying records belonging to a sector they are not authorized to manage.

If access is denied, the user is informed and remains within the sector they were already working in, so that no unauthorized data is exposed. If access is granted, the system loads the data associated with the newly selected sector and refreshes the dashboard accordingly, after which the user

continues working within that sector. This process is what allows a single account, particularly one belonging to the Business Owner, to move across the different areas of the business without needing a separate login for each one.

Overall System Flow Summary

Taken together, the five flowcharts presented in this document describe the core operational cycle of the DYS Sales Tracker Management System, beginning with the process every user must complete before doing anything else — logging in — and continuing through the routine activities of recording sales, recording expenses, reviewing performance through generated reports, and moving between business sectors as needed. Each of these processes corresponds to one of the major features described in the approved Concept Paper: role-based access control, the automated financial calculator, the interactive analytics dashboard, and the business sector switcher.

Documenting these processes as flowcharts before development begins allows the project team to confirm that the behavior expected of the system matches what was proposed to the client, without assuming any functionality beyond what has been approved. The flowcharts also serve as a reference for the diagrams and specifications developed later in the blueprint phase, since the sequence of actions described here is expected to remain consistent with the user flow, the wireframes, and the database structure designed for the system.